Presented by:
Gehan Abdulhameed
Rahma Gamal
Ruwayda Salah
Manar Mohammed
Bassant Adel
Passant Mahmoud

وزارة الاتصالات وتكنولوجيا المعلومات

الرواد الرقميون
DIGILIANS

Under Supervision of:
Dr. Gouda Ismail
Eng. Omar Mamdouh

Head of Track:
Dr. Aya Hossam

AI-Driven Drug Repurposing

A Dual-Model Ensemble Framework Combining DREAMwalk and TxGNN over Biomedical Knowledge Graphs

ABSTRACT

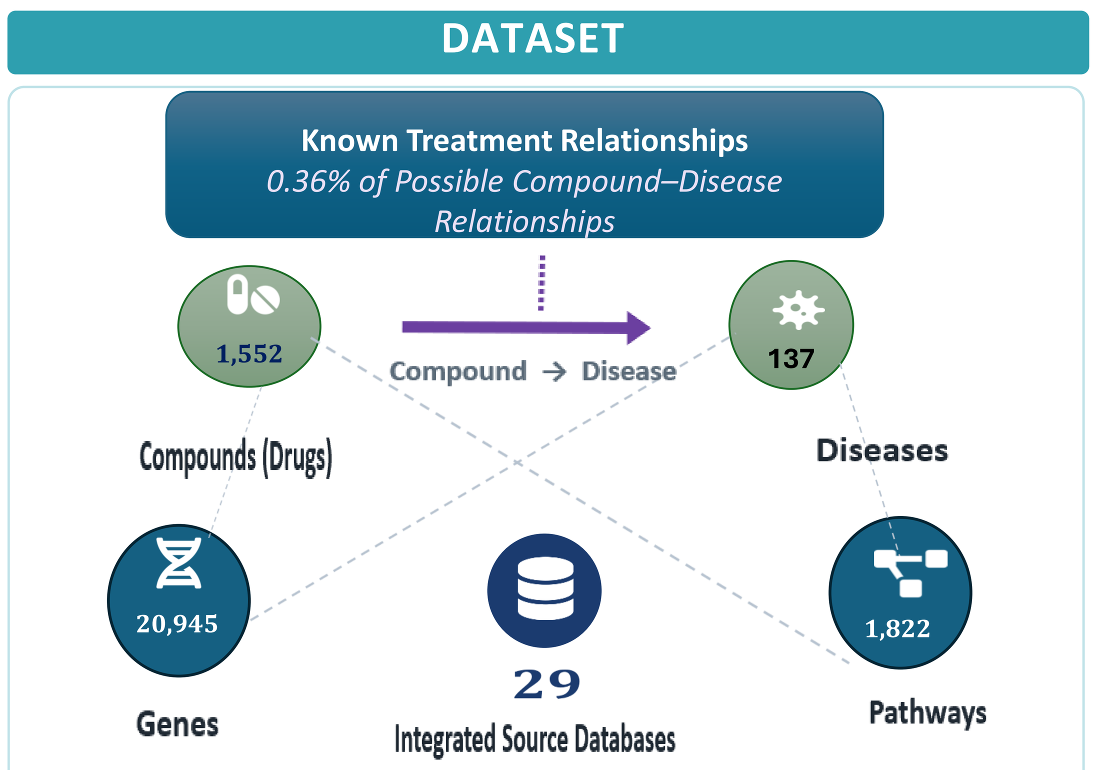
Drug repurposing identifies new therapeutic indications for existing, approved drugs, cutting the cost, time, and clinical risk of conventional drug development. This project implements a dual-model AI framework DREAMwalk (graph embeddings + XGBoost) and TxGNN (a graph neural network) trained independently on the Hetionet biomedical knowledge graph. Their predictions are unified through a weighted ensemble layer that requires agreement between two architecturally different models, reducing false positives and improving robustness on novel, unseen drug–disease associations. SHAP-based explainability makes every prediction traceable to real biological pathways.

AIM OF THE WORK

To design a computational framework that predicts novel, biologically-grounded drug–disease associations by combining two independent AI architectures improving prediction reliability, transparency, and real-world applicability of AI-driven drug repurposing for clinicians and researchers.

SYSTEM OVERVIEW

DATASET



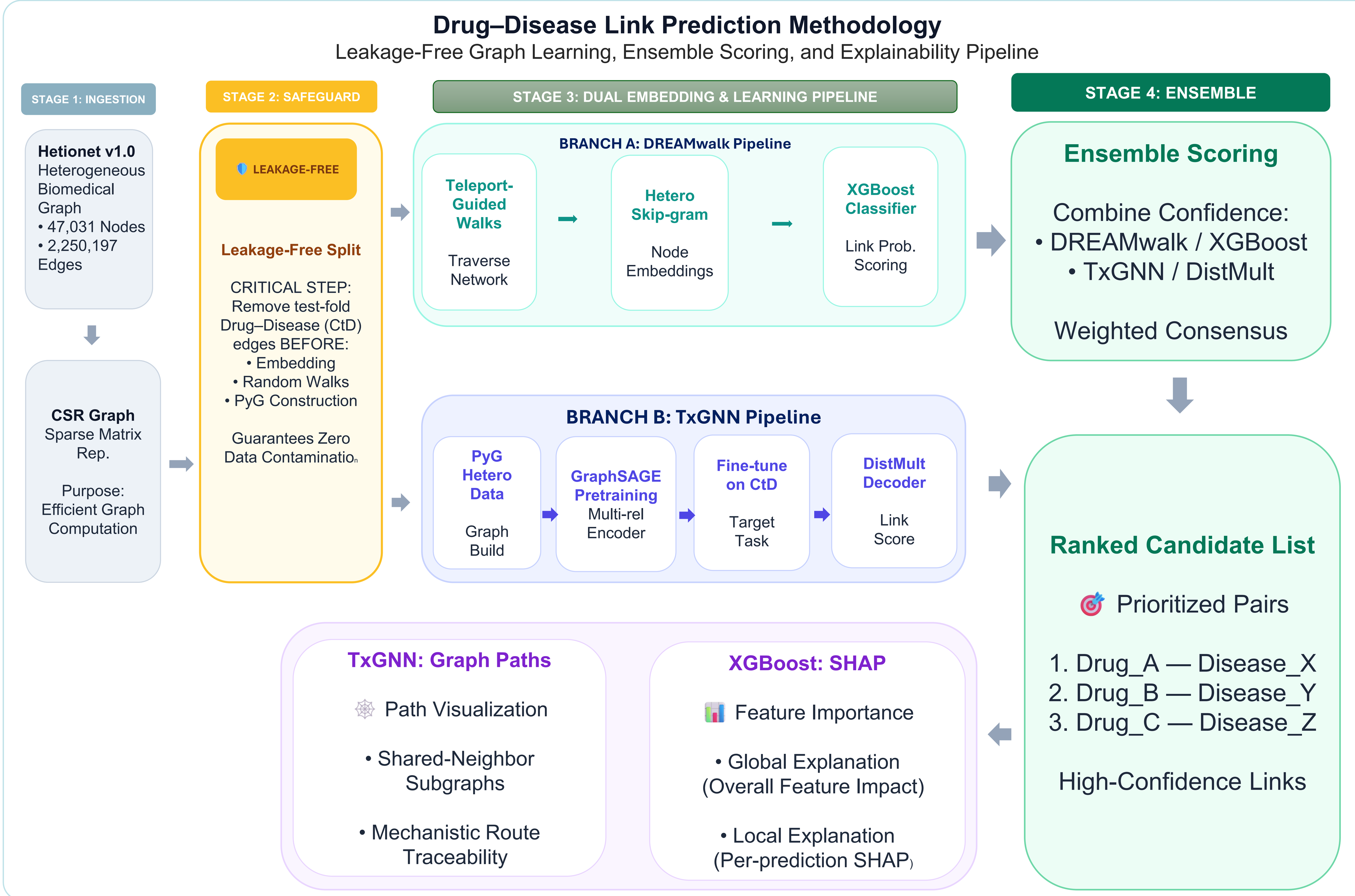
USED TOOLS

GENSIM
scikit learn
PyTorch
Shap
SciPy / NumPy
XGBoost
PyG (Geometric)

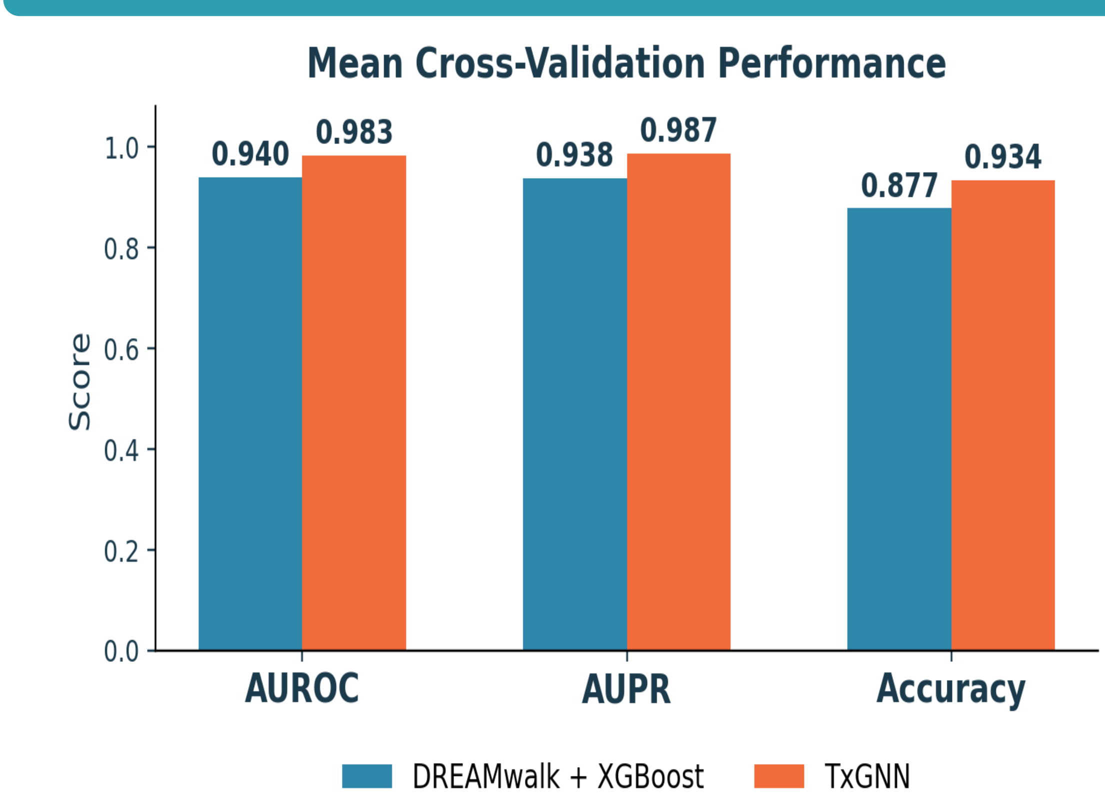
KEY REFERENCES

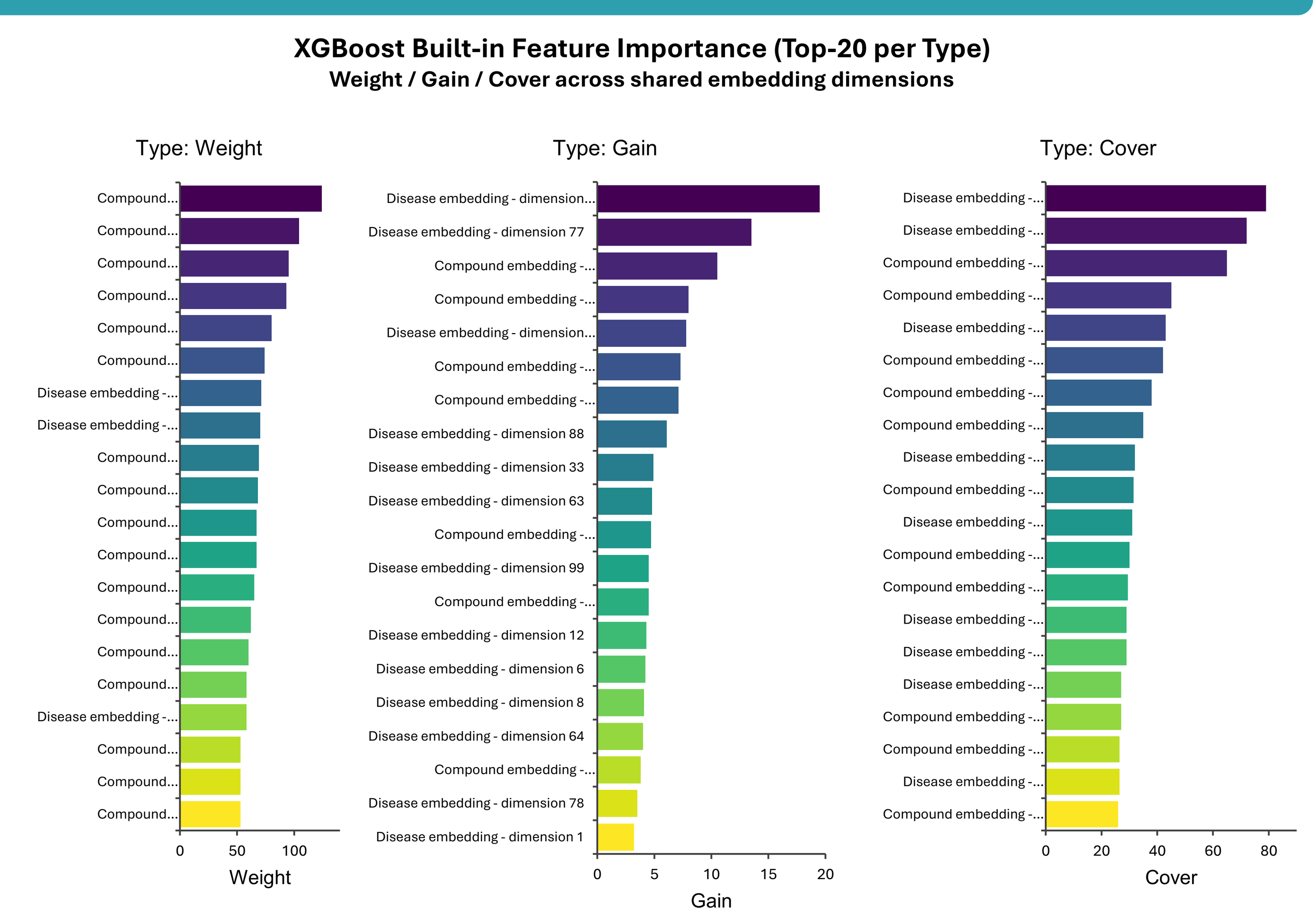
- Bang, Lim, Lee & Kim (2023). Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers (DREAMwalk). Nature Communications.
- Wang, Y. et al. (2022). A novel approach to repurposing drugs based on chemical and genomic features. GCN + GCMC + Attention Fusion.
- Huang, K. & Zitnik, M. (2024). A foundation model for clinician-centered drug repurposing (TxGNN).
- Amiri, Razmara, Parvizpour & Izadkhah (2023). A novel efficient drug repurposing framework via drug–disease association data integration using CNNs.

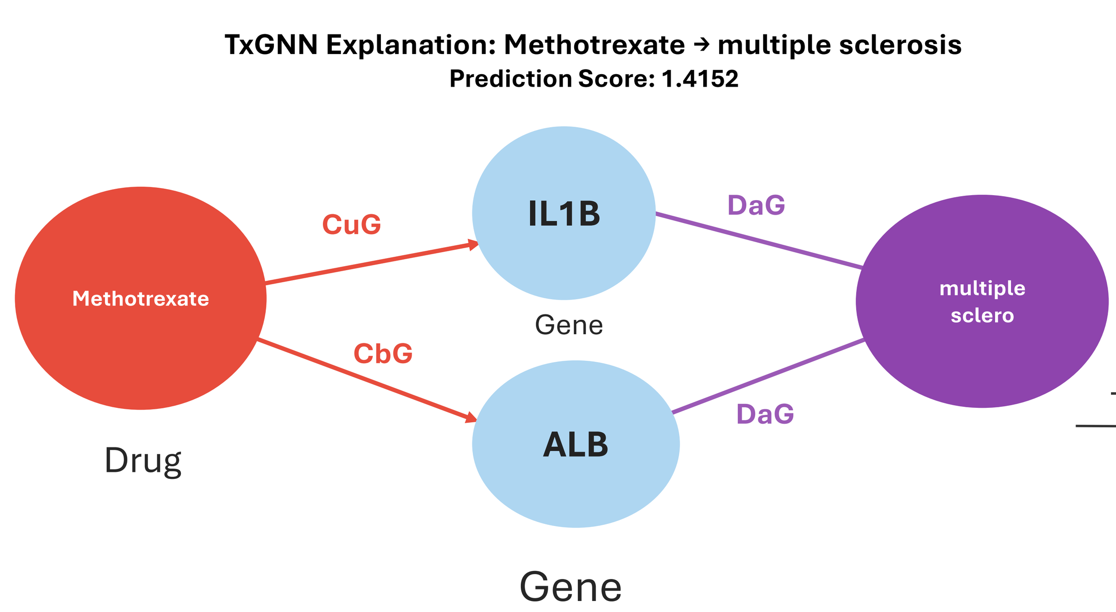
METHODOLOGY

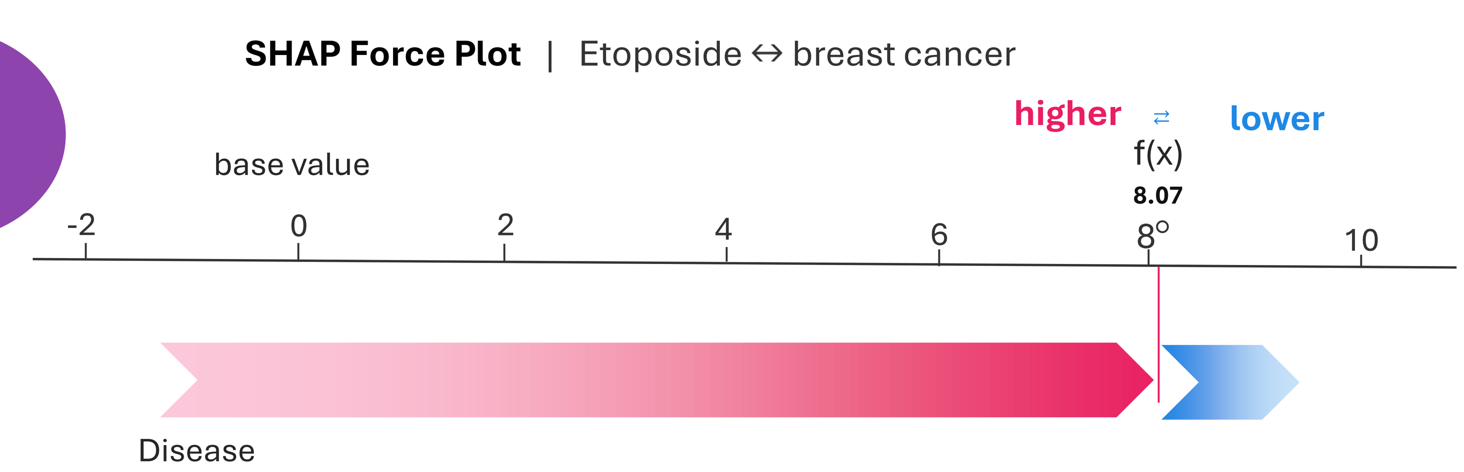


RESULTS









FUTURE WORK

- Extend evaluation to additional knowledge graphs (MSI, KEGG) for cross-KG validation.
- Incorporate multi-omics data (genomic, transcriptomic, proteomic) for richer node features.
- Pursue in-vitro / in-vivo validation of top-ranked repurposing candidates.

CONCLUSION

Combining DREAMwalk and TxGNN into a single ensemble delivers more reliable, explainable drug-repurposing predictions than either model alone. By enforcing a leakage-free evaluation protocol and integrating SHAP-based explainability, the framework directly addresses the data incompleteness, black-box, and generalizability challenges that limit current AI-driven drug repurposing systems — moving the field closer to clinically trustworthy hypothesis generation.

DEMO & PRESENTATION

Scan to explore the interactive demo: query a disease, view ranked repurposing candidates, and inspect SHAP-based explanations through the XAI assistant chatbot.

Scan Me